THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET 03

Full Syllabus (Chapters 1–14) | Class 9 Maths | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(b)** $(4, -4)$	Quadrant II needs $x < 0,\ y > 0$
2	**(a)** 1 and 3	compare with $y = ax + b$
3	**(b)** rational	repeating decimal $= \frac{1}{3}$
4	**(c)** $25x^2 - 10x + 1$	$(a-b)^2 = a^2 - 2ab + b^2$
5	**(d)** 90°	angle in a semicircle
6	**(a)** 115°	opposite angles of a cyclic quadrilateral add to 180°
7	**(a)** 154 cm^2	$\pi r^2 = \frac{22}{7} \times 7^2 = 154$
8	**(a)** 0.15	$6/40 = 0.15$
9	**(d)** 1	always happens
10	**(d)** 55	$1 + 2 + \cdots + 10$
11	**(d)** right-angled, since $a^2 + b^2 = c^2$	$49 + 576 = 625$ and $25^2 = 625$
12	**(b)** 46	groups of equal size, so add the averages and divide by their number
13	**(a)** the two groups have different sizes	the larger group should count for more
14	**(a)** the largest possible column total is 9 + 9 + 1 = 19	19 = 9 + 10 so carry is at most 1
15	**(d)** 143	repeated reduction ends at gcd(143, 0)
16	**(a)** 13 cm	PQ = BC/2
17	**(a)** a square	equal diagonals ⇒ rhombus; perpendicular diagonals ⇒ rectangle
18	**(b)** parallel to the y-axis	x stays 5 while y varies
19	**(d)** $2r$	$2\pi r \times 2r = 4\pi r^2$
20	**(c)** 360 cm^3	$lwh = 15 \times 6 \times 4 = 360$

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	$(0, 6)$ and $(-4, 0)$; distance $= \sqrt{16 + 36} = \sqrt{52} = 2\sqrt{13}$ units.	1+1
22	$2n - 1$. $2n - 1 = 47 \Rightarrow n = 24$.	1+1
23	$4 \times 15 = 60$ and $6 \times 10 = 60$, so $ad = bc$ ⇒ equal. (Both equal $\frac{2}{3}$.)	1+1
24	Height $= \sqrt{25 - 9} = 4$ cm; area $= \frac{1}{2} \times 6 \times 4 = 12$ cm^2.	1+1
25	A proposition claims truth for all cases (1). One case where the ‘if’ part holds but the ‘then’ part fails contradicts the claim, e.g. 9 for “every odd number is prime” (1).	1+1
26	Profit $= 0.20 \times 2000 + 0.30 \times 3000 = 400 + 900 = 1300$ (1); percentage $= \frac{1300}{5000} \times 100 = \mathbf{26\%}$ (1).	1+1
27	For 6 km: $x + 6y = 110$ (1). For 10 km: $x + 10y = 170$ (1).	1+1
28	Rotate the triangle through 360° about one of its two perpendicular sides (1); this side becomes the height, the other becomes the radius and the hypotenuse the slant height (1).	1+1

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	Sides: 4, 3 and $\sqrt{16 + 9} = 5$. Perimeter $= 12$ units.	2+1
30	Let MN meet AD at X (1). $MN \parallel BC$ (Midpoint Theorem), so in $\triangle ABD$ the line MX through the midpoint M of AB is parallel to BD (1). By the converse of the Midpoint Theorem X is the midpoint of AD (1).	1+1+1
31	(i) rational, 9. (ii) irrational. (iii) rational, $\frac{1}{3}$. (iv) rational (repeating block 12345): $\frac{12345}{99999} = \frac{4115}{33333}$. Marking: ½+½+1+1.	½+½+1+1
32	A line meets a circle in at most two points; hence three points on a line cannot all be on one circle (equivalently, a circle can be drawn through three points only if they are non-collinear).	1+2

33	$S = 1 + 2 + \cdots + n$ and $S = n + (n-1) + \cdots + 1$ (1). Adding: $2S = n(n+1)$ since there are n pairs, each $n+1$ (1). So $S = \frac{n(n+1)}{2}$ (1).	1+1+1
34	Slope of $l_1 = \dfrac{8-2}{3-0} = 2$ (1). Slope of $l_2 = \dfrac{0-(-4)}{2-0} = 2$ (1). Equal slopes with different y-intercepts (2 and -4): parallel lines, so no solution (1).	1+1+1
35	Annual marks: weights are the importance of internals, project, exam (1½). Mixtures/alloys: weights are the quantities mixed (1½).	1½+1½
	OR: Equal sizes ⇒ simple average (1½): $\frac{2+3+4+3}{4} = 3$ hours (1½).	1½+1½
36	Circumference $= 176$ cm (1). $1.76\ \text{km} = 176000$ cm (1). Revolutions $= \frac{176000}{176} = 1000$ (1).	1+1+1
	OR: Arc $= \frac{60}{360} \times 2 \times \frac{22}{7} \times 21 = 22$ cm (2). Perimeter $= 22 + 21 + 21 = 64$ cm (1).	2+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) $2x + 3y = 80$, $3x + 2y = 70$ (1). (ii) Adding: $5x + 5y = 150$; subtracting: $x - y = -10$, so $x + y = 30$ and $x - y = -10$ give $x = 10$, $y = 20$ (2). (iii) ₹30 (1). (iv) Exactly one pair of prices fits both bills; the lines intersect at one point (1).	1+2+1+1
	OR: Let (x_1, y_1) and (x_2, y_2) lie on the line: $y_1 = mx_1 + d$ and $y_2 = mx_2 + d$ (1). Slope $= \dfrac{y_2 - y_1}{x_2 - x_1} = \dfrac{m(x_2 - x_1)}{x_2 - x_1} = m$ (2). Rewrite $-3y = -6x - 9$, so $y = 2x + 3$ (1). Slope 2, y-intercept 3 (1).	1+2+1+1
38	(i) If my profit is more, then I have sold more articles (1). (ii) 10 pens × ₹4 = ₹40; 20 pens × ₹1 = ₹20; more pens, less profit (1½). (iii) 5 laptops at ₹2000 profit = ₹10000 versus 50 pens at ₹10 profit = ₹500: more profit with fewer articles sold (1½). (iv) Counterexamples (1).	1+1½+1½+1
	OR: (i) 15: 1, 3, 5, 15; 33: 1, 3, 11, 33 — 4 divisors each (2). (ii) Converse: If n has exactly 4 divisors, then it is a product of two unequal primes (1). False: 8 has divisors 1, 2, 4, 8 (four divisors) but $8 = 2^3$ is not a product of two unequal primes (2).	2+3
39	$x^2 + \frac{1}{x^2} = 25 - 2 = 23$. $x^3 + \frac{1}{x^3} = \left(x + \frac{1}{x}\right)^3 - 3\left(x + \frac{1}{x}\right) = 125 - 15 = 110$.	2+3
	OR: (i) $(x-2)(x-3)$; (ii) $(r-7)(r+6)$; (iii) $(2x+1)(3x+2)$; (iv) $(2x-3)(5x+2)$ [since $10x^2 - 15x + 4x - 6$].	1+1+1+2
40	(i) $\frac{10}{50} = 0.2$ (1). (ii) bus 0.36, bicycle 0.2, walking 0.24, car 0.2; sum $= 1$ (2). (iii) $0.24 \times 1200 = 288$ (1). (iv) Bus is most likely (0.36); least likely: bicycle/car (0.2), difference 0.16 (1).	1+2+1+1
	OR: (i) $\frac{8}{200} = 0.04$ (1). (ii) $0.04 \times 5000 = 200$ (1). (iii) 0.96 (1). (iv) A bigger, representative sample varies less from the true proportion (2).	1+1+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.